

CPEN 416

Lecture 05: Multi-qubit states and gates

Thursday 24 September 2026

Announcements

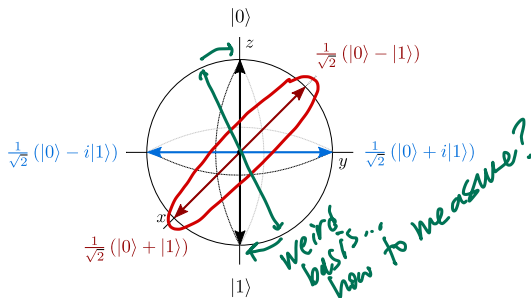
- Monday: tutorial assignment 1 (quantum key distribution)
- Quiz 3 on Tuesday
- Midterm in class on Tuesday 6 Oct - details and past midterms available on the weekend in PrairieLearn

Last time

We formalized the idea of *projective measurements*: if we measure a qubit in state $|\varphi\rangle$ with respect to basis $\{|\psi_i\rangle\}$, the probability of observing it in state $|\psi_i\rangle$ (outcome i) is

$$\Pr(\text{outcome } i) = |\langle \psi_i | \varphi \rangle|^2$$

Visually: geometric projection on the axis represented by a basis.



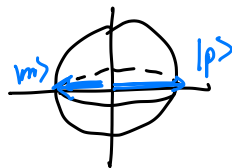
Last time

To measure in alternative bases, must apply a basis rotation.

Example: measure in Y basis. First, determine mapping between computational basis and other basis:

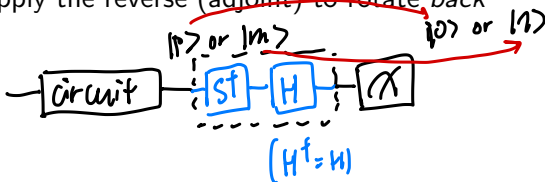
$$|0\rangle \rightarrow |p\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$$

$$|1\rangle \rightarrow |m\rangle = \frac{1}{\sqrt{2}} (|0\rangle - i|1\rangle)$$



At the end of our circuit, apply the reverse (adjoint) to rotate back to the computational basis.

$$\begin{aligned} |0\rangle &\rightarrow [K] \rightarrow [S] \rightarrow |p\rangle \\ |1\rangle &\rightarrow [K] \rightarrow [S] \rightarrow |m\rangle \end{aligned}$$



Last time

We ran the following circuit and measured in the Y basis.



*Weekend fun: derive
Probs. as a function of θ*

Learning outcomes

- Mathematically describe a system of multiple qubits
- Describe the action of common multi-qubit gates
- Apply gates controlled on the state of multiple qubits
- Define and give examples of entangled states

Tensor products

Hilbert spaces compose under the *tensor product*.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}.$$

The tensor product of A and B , $A \otimes B$ is

$$A \otimes B = \begin{pmatrix} a \begin{pmatrix} e & f \\ g & h \end{pmatrix} & b \begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ c \begin{pmatrix} e & f \\ g & h \end{pmatrix} & d \begin{pmatrix} e & f \\ g & h \end{pmatrix} \end{pmatrix} = \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

Multi-qubit states

Qubit state vectors also combine under the *tensor product*:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Exercise: Determine the 2-qubit computational basis.

$$|0\rangle \otimes |0\rangle = |00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |0\rangle \otimes |1\rangle = |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |1\rangle \otimes |0\rangle = |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |1\rangle \otimes |1\rangle = |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$|0010\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \updownarrow 16$$

$$|1101\rangle \rightarrow 13$$

$$2^3 + 2^2 + \dots$$

Question: How many n -qubit basis states? (dimension of space?) 2^n

exponentially large!!

Multi-qubit states

The tensor product is linear and distributive. Given

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\varphi\rangle = \gamma|0\rangle + \delta|1\rangle,$$

$$|\psi\rangle \otimes |\varphi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes \begin{pmatrix} \gamma \\ \delta \end{pmatrix} = \dots$$

$$= (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$$

$$= \alpha\gamma|00\rangle + \beta\gamma|10\rangle + \alpha\delta|01\rangle + \beta\delta|11\rangle$$

$$= \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}$$

Apply $X \otimes Z$: $|\psi\rangle \rightarrow \beta|0\rangle + \alpha|1\rangle$
 $|\varphi\rangle \rightarrow \gamma|0\rangle - \delta|1\rangle$

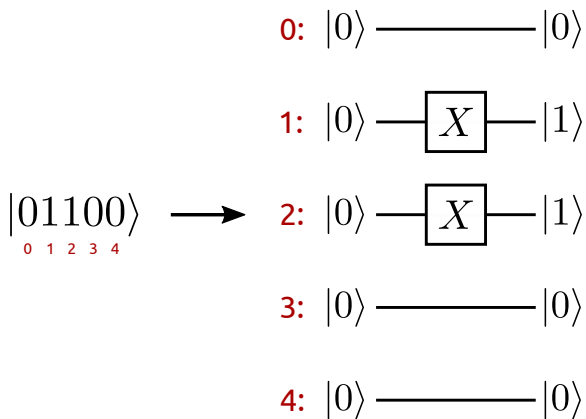
$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix} = \begin{pmatrix} \beta\gamma \\ -\beta\delta \\ \alpha\gamma \\ -\alpha\delta \end{pmatrix}$$

Question: Are tensor products of states still normalized? (yes)

check: $|\alpha\gamma|^2 + |\alpha\delta|^2 + |\beta\gamma|^2 + |\beta\delta|^2 = 1$

Qubit ordering (very important!)

In PennyLane:

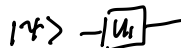


This is different in other frameworks.

Multi-qubit gates

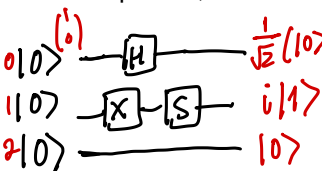
Unitary operations also compose under tensor product.

Example: apply U_1 to $|\psi\rangle$ and U_2 to $|\varphi\rangle$



$$(U_1|\psi\rangle) \otimes (U_2|\varphi\rangle) = (U_1 \otimes U_2) (|\psi\rangle \otimes |\varphi\rangle)$$

Exercise: determine the state of a 3-qubit system if H is applied to qubit 0, X and then S are applied to qubit 1.



$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}|1\rangle$$

$$i|1\rangle$$

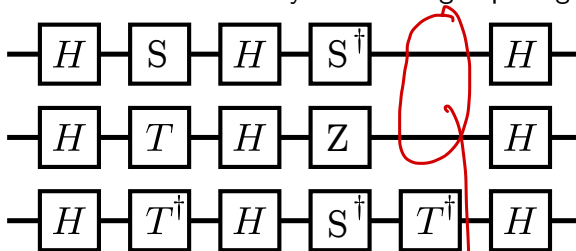
$$|0\rangle$$

- How many basis states involved? 2
- How many entries in the vector? 2
- Express using ket notation vs. vector
- Test it's normalized.

$$\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) \otimes i|1\rangle \otimes |0\rangle = \frac{i}{\sqrt{2}} (|010\rangle + |110\rangle) = \frac{i}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

Multi-qubit gates

The circuits we've seen so far only involve single-qubit gates:



Surely this isn't all we can do...

multi-qubit operations.

Image credit: Xanadu Quantum Codebook I.11

SWAP

Define an operation that swaps the state of two qubits. First, specify action on basis states:

$$|00\rangle \rightarrow |00\rangle$$

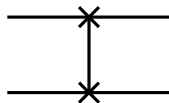
$$|01\rangle \rightarrow |10\rangle$$

$$|10\rangle \rightarrow |01\rangle$$

$$|11\rangle \rightarrow |11\rangle$$

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Circuit element:



PennyLane: `qp.SWAP([0, 1])`

$$\begin{aligned} & |0001\rangle \\ & |+-+\rangle \\ & \downarrow \\ & |+\rangle \otimes |-\rangle \otimes |+\rangle \\ & = \frac{1}{\sqrt{2}} (|000\rangle + |100\rangle + \dots) \end{aligned}$$

CNOT

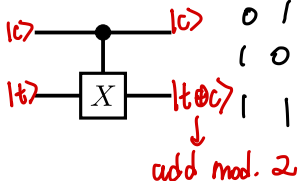
CNOT = “controlled-NOT”. A NOT (X) is applied to second qubit only if first qubit is in state $|1\rangle$.

$$\begin{aligned}
 |00\rangle &\rightarrow |00\rangle \\
 |01\rangle &\rightarrow |01\rangle \\
 |10\rangle &\rightarrow |11\rangle \\
 |11\rangle &\rightarrow |10\rangle
 \end{aligned}$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

c t		c t	
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

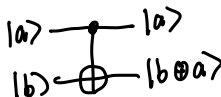
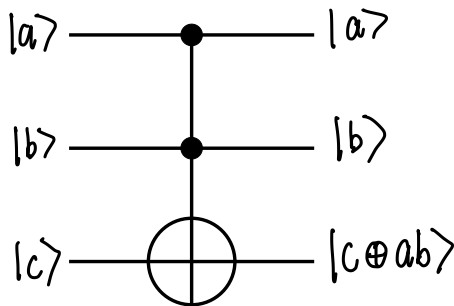
Circuit elements:



PennyLane: `qp.CNOT`

Toffoli (TOF)

There are also gates on more than two qubits, like the Toffoli gate, which is a controlled-CNOT, or controlled-controlled-NOT.



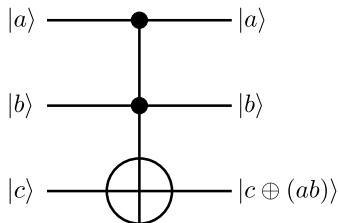
$$\begin{aligned} \text{TOF}|000\rangle &= \\ \text{TOF}|001\rangle &= \\ \text{TOF}|010\rangle &= \\ \text{TOF}|011\rangle &= \\ \text{TOF}|100\rangle &= \\ \text{TOF}|101\rangle &= \end{aligned} \left. \vphantom{\begin{aligned} \text{TOF}|000\rangle &= \\ \text{TOF}|001\rangle &= \\ \text{TOF}|010\rangle &= \\ \text{TOF}|011\rangle &= \\ \text{TOF}|100\rangle &= \\ \text{TOF}|101\rangle &= \end{aligned}} \right\} \text{same.}$$
$$\begin{aligned} \text{TOF}|110\rangle &= |111\rangle \\ \text{TOF}|111\rangle &= |110\rangle \end{aligned}$$

PennyLane: `qp.Toffoli`

Quantum circuits and Boolean functions

The Toffoli implements a reversible AND gate. It is also universal for classical reversible computing.

a	b	$a \wedge b$
0	0	0
0	1	0
1	0	0
1	1	1

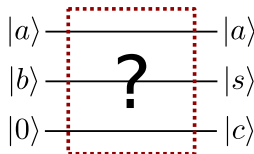


Exercise: what Boolean function does CNOT implement? (XOR)

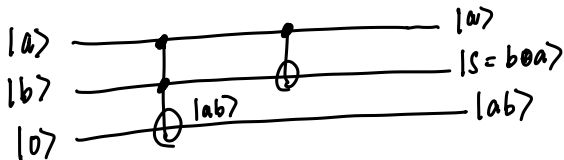
Quantum circuits and Boolean functions

X, CNOT, TOF can be used to create Boolean arithmetic circuits.

Exercise: use a combination of these gates to create a quantum version of a one-bit adder. Test it in PennyLane on all possible bit values of a and b .



*challenge:
make ans
of gates*



Next time

Content:

- Multi-qubit measurement and the Bell basis
- Superdense coding

Action items:

- ➊ Prepare for Quiz 3
- ➋ Work on practice assignment 1
- ➌ Work through exercises from tutorials

Recommended reading:

- Codebook nodes IQC, SQ, MQ; Nielsen & Chuang 1.2, 1.3.1-1.3.7, 2.2.3-2.2.5, 2.3, 4.3